magnify in order to make them easily visible. At first as we examine the fabric of space on shorter distance scales, not much happens; it appears to retain the same basic form as it has on larger scales, as we see in the first three levels of magnification. However, as we continue on our journey toward the most microscopic examination of space—the fourth level of magnification in Figure 8.3—a new, curled-up, circular dimension becomes apparent, much like the circular loops of thread making up the pile of a tightly woven piece of carpet. Kaluza and Klein suggested that the extra circular dimension exists at every point in the extended dimensions, just as the circular girth of the garden hose exists at every point along its unfurled, horizontal extent. (For visual clarity, we have drawn only an illustrative sample of the circular dimension at regularly spaced points in the extended dimensions.) We show a close-up of the Kaluza-Klein vision of the microscopic structure of the spatial fabric in Figure 8.4.

The similarity with the garden hose is manifest, although there are some important differences. The universe has three large, extended space dimensions (only two of which we have actually drawn), compared with the garden hose's one, and, more important, we are now describing the spatial fabric of the universe itself, not just an object, like the garden hose, that exists within the universe. But the basic idea is the same: Like the circular girth of the garden hose, if the additional curled-up, circular dimension of the universe is extremely small, it is much harder to detect than the manifest, large, extended dimensions. In fact, if its size is small enough, it will be beyond detection by even our most powerful magnifying instruments. And, of utmost importance, the circular dimension is not merely a circular bump within the familiar extended dimensions as the illustration might lead you to believe. Rather, the circular dimension is a new dimension, one that exists at every point in the familiar extended dimensions just as each of the up-down, left-right, and back-forth dimensions exists at every point as well. It is a new and independent direction in which an ant, if it were small enough, could move. To specify the spatial location of such a microscopic ant, we would need to say where it is in the three familiar extended dimensions (represented by the grid) and also where it is in the circular dimension. We would need four pieces